

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4301

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems.(L4,L5)

Assessment Tool : Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?

17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I CH.Bhavya Sri/192411249 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

CH.Bhavya Sri

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

SOAP (Simple Object Access Protocol) is an XML-based protocol used for exchanging structured information between applications over a network.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to describe a web service, its operations, messages, data types, and endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules and interfaces that allows one software application to communicate with another.

4. Define XML Schema.

XML Schema (XSD) defines the structure, elements, attributes, and data types allowed in an XML document and is used to validate XML data.

5. What is a Web Service?

A **Web Service** is a software service that allows different applications to communicate and exchange data over a network using standard web technologies.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services using resources and standard HTTP methods such as GET, POST, PUT, and DELETE.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is formatted, transmitted, and received between communicating devices or applications.

8. Define Endpoint in web services.

An **endpoint** is the specific URL or network address through which a web service can be accessed.

Example: <https://example.com/api/users>

9. What is a Service in web applications?

A **service** is a software component that provides a specific function or functionality to other applications through a defined interface.

10. What is the difference between SOAP and REST?

SOAP	REST
Protocol	Architectural style
Uses XML messages	Commonly uses JSON
More structured and formal	Lightweight and simpler
Higher processing overhead	Generally lower overhead
Supports WS-* standards	Commonly uses HTTP security mechanisms

11. What is the role of WSDL in web services?

WSDL acts as a **contract** between a web service provider and its client. It describes the available operations, input/output messages, data types, bindings, and service endpoint.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight data-interchange format. APIs commonly use JSON to send structured data between clients and servers.

Example:

```
{  
  "name": "Ravi",  
  "age": 25  
}
```

13. What is HTTP and why is it used in web services?

HTTP (Hypertext Transfer Protocol) is a communication protocol used to transfer data between clients and servers. Web services commonly use HTTP because it supports standard request methods such as GET, POST, PUT, and DELETE.

14. What is meant by request and response in web communication?

A **request** is a message sent by a client to a server asking for a resource or operation.

A **response** is the message returned by the server containing the requested data or the result of the operation.

Client → Request → Server

Client ← Response ← Server

15. What is the importance of XML in web services?

XML provides a **structured, platform-independent format** for representing and exchanging data. It is especially important in SOAP-based web services.

16. What is the purpose of an API endpoint?

An **API endpoint** provides a specific address through which a client can access a particular resource or operation of an API.

17. What is the difference between HTTP and HTTPS?

HTTP transfers data without TLS encryption.

HTTPS (HTTP Secure) uses **TLS encryption** to protect data during transmission and also helps authenticate the server.

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They use resources identified by URLs and standard HTTP methods to perform operations on those resources.

19. What is a namespace in XML?

An **XML namespace** uniquely identifies XML elements and prevents naming conflicts when elements from different XML vocabularies are used together.

Example:

`xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"`

20. What is the role of a client and server in web services?

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Client: Sends requests to access a service or resource.

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Server: Receives the request, processes it, and sends a response.

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Request

Client -----> Server

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Response